

Cross-Circuit Formula 1 Telemetry Performance Decomposition

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*Research artifact generated from the RaceCast telemetry analysis pipeline
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Abstract

This study evaluates a public-telemetry workflow for decomposing Formula 1 qualifying-lap performance into local track windows and tests whether machine learning adds value beyond a direct analytic calculation. Data from five 2024 qualifying sessions were filtered using explicit quality rules and converted into 372 same-driver lap comparisons and 2,487 adaptive segment observations. The original gradient-boosting model achieved mean leave-one-circuit-out MAE of 0.0563 s and mean R^2 of 0.7258. However, a non-learned inverse-speed reconstruction derived from segment length, reference segment time, and mean-speed delta performed better, with mean MAE of 0.0408 s and mean R^2 of 0.8127. After subtracting the analytic reconstruction, a control/powertrain model failed to predict the residual on every held-out circuit (mean $R^2 = -0.275$). These results show that the initial machine-learning formulation largely approximated a relationship already available analytically from speed and distance, and that the evaluated public control channels did not provide transferable residual predictive signal. The contribution is therefore methodological rather than a claim that machine learning discovers independent causes of lap-time loss: the work documents an open telemetry-decomposition pipeline, demonstrates the importance of strong analytic baselines, and identifies failure modes involving target coupling, scale-dependent evaluation, adaptive segmentation, and public-data limitations.

Keywords: Formula 1; telemetry; lap-time decomposition; motorsport analytics; analytic baseline; machine learning; cross-circuit validation; negative result.

1. Engineering Motivation and Research Objectives

An overall lap-time delta answers how much faster or slower one lap is, but not where the difference accumulated or whether a learned model is necessary to explain it. RaceCast was developed to transform public telemetry into localized comparisons of speed, braking, throttle, RPM, gear and other available channels. The initial research question asked whether segment-level time loss could be predicted on a circuit excluded from training. A methodological audit revealed that the target - traversal-time difference - is mathematically linked to speed-derived predictors. The revised paper therefore asks a stricter question: does machine learning add transferable information beyond the analytic speed-time relationship already present in the telemetry?

The revised research questions are:

- RQ1: How accurately can a direct analytic inverse-speed reconstruction reproduce segment-level time differences across held-out circuits?
- RQ2: Do the evaluated learned models outperform simple non-learned baselines under the same circuit-held-out evaluation?
- RQ3: After removing the analytic speed-based component of the target, do public control and powertrain features predict the remaining residual across circuits?
- RQ4: How do target scale, adaptive segmentation, selection rules and public-telemetry limitations affect interpretation?

2. Data and Selection

The dataset was extracted through FastF1 3.8.3 from 2024 Formula 1 qualifying sessions at Bahrain, Monaco, Silverstone, Monza and Suzuka. Candidate laps were required to have a valid lap time, pass FastF1's `IsAccurate` flag when available, have no pit-in or pit-out marker, and satisfy FastF1 `pick_quicklaps` with a 1.07 threshold. No additional outlier trimming was applied. This rule produced 469 selected laps. Telemetry extraction succeeded for 467; one Monaco lap (ALB lap 28) and one Suzuka lap (MAG lap 2) failed because telemetry processing returned an empty sequence. The final comparison set contains 372 same-driver lap pairs.

Circuit	Selected laps	Drivers	Pairs
Bahrain	85	20	65
Monaco	143	20	122
Monza	87	20	67
Silverstone	80	15	65
Suzuka	74	20	53

The 1.07 quick-lap rule resulted in only 15 represented drivers at Silverstone, compared with 20 at the other events. The paper reports this as a consequence of the filter rather than inferring a causal reason. Across all pairs, 98.9% used the same tyre compound. Official lap-time deltas had median 0.661 s, 95th percentile 3.228 s and maximum 5.566 s, so the dataset includes substantially compromised or non-reference-like laps; this is treated as a limitation rather than silently removed.

3. Telemetry Alignment and Adaptive Segmentation

Telemetry was aligned to a 5 m distance grid. The compact file named `delta_time_50m.csv` stores every tenth aligned sample for visualization, so its exported spacing is 50 m; the analysis itself uses 5 m. Local traversal time was approximated as $\Delta d/v$ after converting speed to m/s, and cumulative comparison-to-reference time delta was obtained by summing local traversal-time differences.

For each driver and event, the fastest selected telemetry-available lap was used as the reference. Local windows were detected from speed minima on that driver's reference lap. This is an adaptive segmentation scheme, not a canonical corner map. Bahrain, Monaco and Monza produced stable segment counts and small center dispersion across drivers, while Silverstone and Suzuka did not: maximum segment-center standard deviation reached approximately 286 m and 519 m, respectively. Accordingly, this revision does not interpret a shared `segment_id` as the same corner across drivers at those circuits. Each row is treated as a local performance window, and fixed circuit-level segmentation is reserved for future work.

Circuit	Segments per driver	Max center SD (m)
Bahrain	8	8
Monaco	7	8
Monza	6	11
Silverstone	5,6	286
Suzuka	6,7	519

4. Models, Baselines and Validation

The original primary analysis compared linear regression, random forest regression and histogram gradient boosting. The configured gradient-boosting model used `learning_rate=0.06`, `max_iter=350`, `max_leaf_nodes=31`, `l2_regularization=0.5` and `random_state=42`. Evaluation used leave-one-circuit-out validation: all observations from one circuit were withheld, the model was trained on the other four, and the process was repeated for all five circuits. The original feature set included speed, braking, throttle, exit-acceleration, RPM, gear, DRS, segment length, reference segment time, and tyre-life delta in the headline model. The earlier audit table omitted tyre-life delta; this discrepancy is now disclosed rather than presented as the same run.

The revised analysis adds three mandatory non-learned baselines: predicting zero, predicting the training-target mean, and an analytic inverse-speed reconstruction. The analytic estimate is:

$$v_{\text{ref}} = L / t_{\text{ref}}; \quad v_{\text{cmp}} \approx v_{\text{ref}} + \Delta v_{\text{mean}}; \quad \Delta t_{\text{analytic}} = L / v_{\text{cmp}} - t_{\text{ref}}$$

where L is segment length, t_{ref} is reference segment time, and Δv_{mean} is mean-speed difference converted to m/s. This formula requires no fitted parameters.

5. Results

5.1 Learned models versus baselines

Method	Mean MAE (s)	Fold SD MAE	Mean R ²
Analytic reconstruction	0.0408	0.0096	0.8127
Gradient Boosting	0.0563	0.0310	0.7258
Random Forest	0.0575	0.0335	0.6965
Linear Regression	0.0701	0.0373	0.6261
Predict training mean	0.1330	0.0611	-0.2975
Predict zero	0.1570	0.0890	-0.7129

The analytic reconstruction is the best method in the evaluated set: mean MAE 0.0408 s and mean R² 0.8127. The original gradient-boosting model is worse at 0.0563 s MAE and 0.7258 mean R². Thus the principal result is negative for the machine-learning hypothesis: the learned model does not earn its complexity when the target can be reconstructed more accurately from the speed-time relationship.

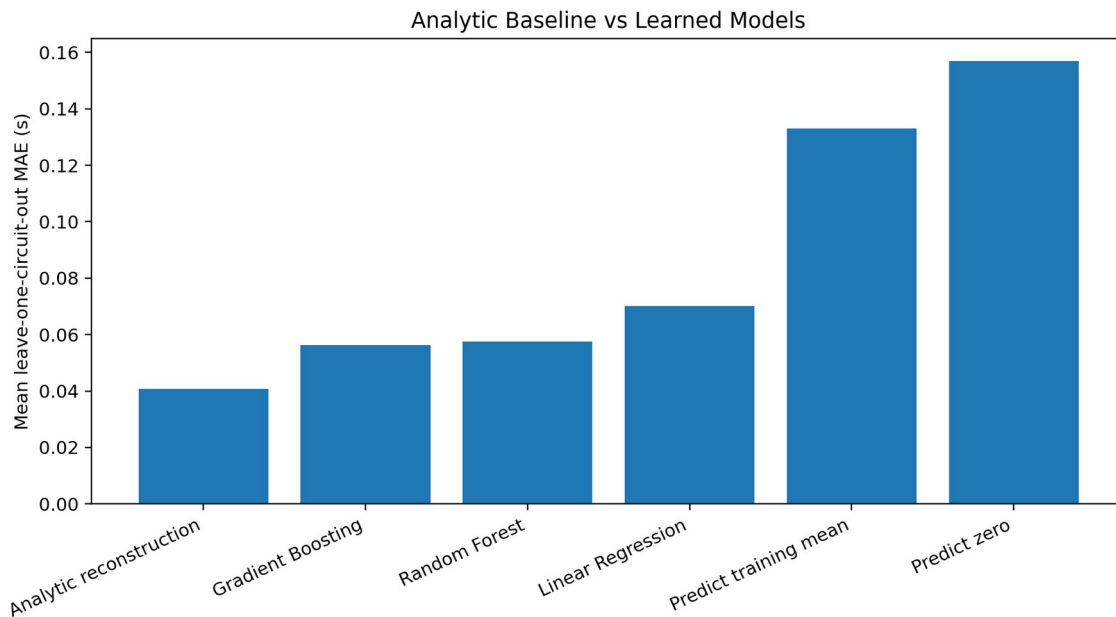


Figure 1. Cross-circuit performance of analytic reconstruction, machine-learning models, and naive reference baselines under leave-one-circuit-out validation, reported as mean absolute error (MAE).

5.2 Circuit-level performance and scale normalization

Circuit	GB MAE	GB R ²	Analytic MAE	Analytic R ²	GB MAE/MAD
Bahrain	0.0344	0.7132	0.0328	0.7520	0.600
Monaco	0.0619	0.6765	0.0565	0.7395	0.765
Monza	0.0449	0.7607	0.0395	0.8891	0.839
Silverstone	0.1078	0.7085	0.0419	0.9528	0.735
Suzuka	0.0327	0.7700	0.0333	0.7303	0.540

Raw MAE is scale dependent. Silverstone has the largest raw gradient-boosting MAE (0.1078 s), but its segment-time deltas are also much larger. After normalizing by each circuit's target median absolute deviation, Silverstone is not the worst held-out circuit; Monza has the highest MAE/MAD ratio in this comparison. The earlier interpretation that Silverstone demonstrated uniquely poor transfer is therefore withdrawn.

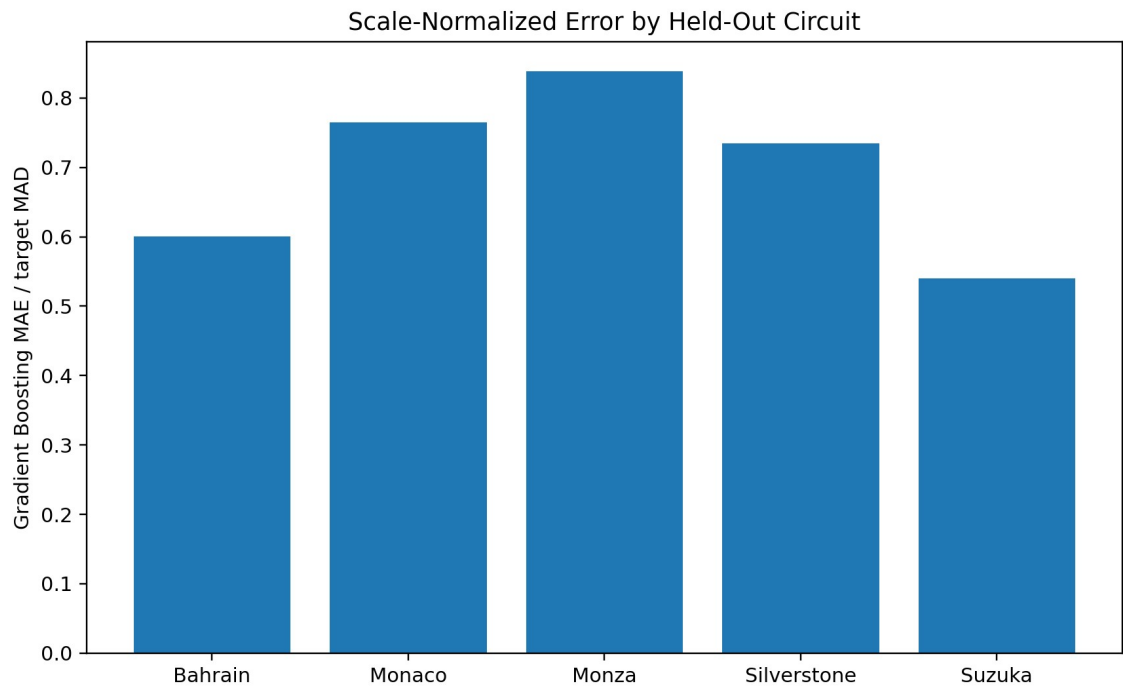


Figure 2. Circuit-specific Gradient Boosting error normalized by the median absolute deviation (MAD) of the segment-time-delta target, providing a scale-adjusted comparison across held-out circuits.

5.3 Residualized experiment

Held-out circuit	n	Residual MAE	Residual R ²
Bahrain	520	0.0360	-0.2770
Monaco	854	0.0575	-0.1038
Monza	402	0.0383	-0.0921
Silverstone	385	0.0567	-0.5906
Suzuka	326	0.0350	-0.3137

After subtracting the analytic reconstruction, a gradient-boosting model using braking, throttle, RPM, gear, DRS, segment length and reference segment time produced negative R² on every held-out circuit; mean R² was -0.275. A negative R² means the model performed worse than predicting the residual mean. Within this public-data formulation, the evaluated control and powertrain features therefore do not provide transferable residual predictive signal beyond the analytic speed-based component.

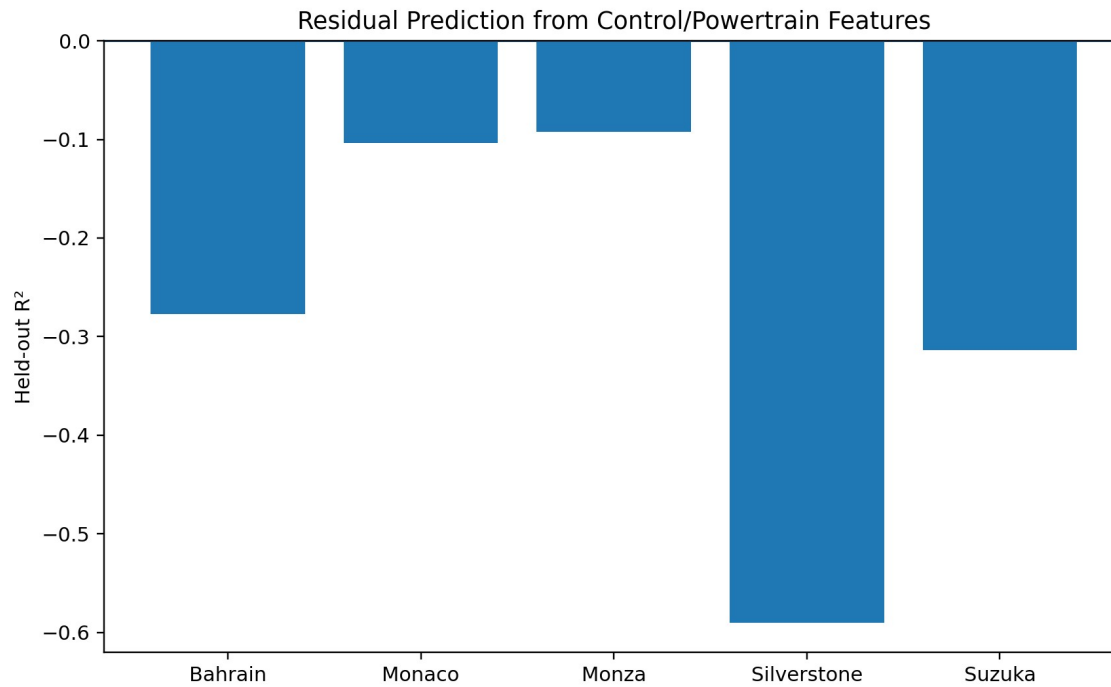


Figure 3. Leave-one-circuit-out coefficient of determination (R^2) for residual segment-time prediction using control and powertrain features after removal of the analytic speed-time component.

5.4 Target distribution, timing fidelity and outliers

The 2,487 segment deltas have mean 0.141 s, median 0.101 s, standard deviation 0.202 s, minimum -0.290 s, 95th percentile 0.495 s, 99th percentile 0.921 s and maximum 2.731 s; 14.23% are negative. The telemetry-integrated lap delta correlates with official lap delta at $r=0.9937$ over the full range, but the correlation falls to $r=0.9428$ when restricting to lap deltas under 1 s. Integration has mean bias +0.0119 s and mean absolute discrepancy 0.0846 s against official lap timing. Because the model predicts a constructed telemetry target rather than official timing, sub-0.1-s model errors should not be interpreted as equivalent physical accuracy against official lap time.

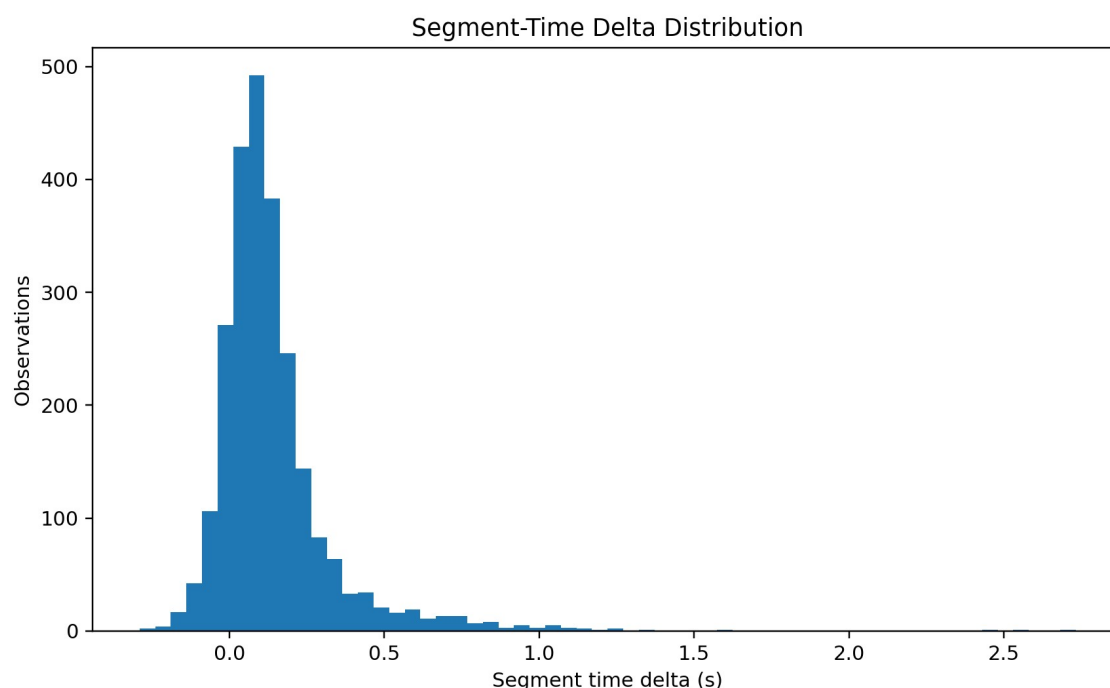


Figure 4. Empirical distribution of constructed segment-level traversal-time differences across the five qualifying-event datasets.

5.5 Feature ablation and explainability

The original ablation table is retained only as descriptive evidence. Gradient boosting fold-to-fold MAE standard deviation is 0.0310 s. Changes from removing braking (-0.00035 s), throttle (-0.00054 s), exit-acceleration (+0.00041 s) or powertrain (-0.00281 s) are far smaller than this cross-circuit variation and are not interpreted as meaningful. Removing the corner-speed group increased mean MAE by 0.0273 s, but this result is expected because speed is algebraically linked to traversal time. SHAP and permutation-importance graphics from the original model are therefore treated as model-behavior diagnostics, not evidence of physical causality or independent feature value.

6. Discussion

The revised result is simpler and stronger than the original story. RaceCast can localize performance differences and organize public telemetry into analyzable windows, but the specific machine-learning target used here does not justify a learned model. A direct inverse-speed calculation is both simpler and more accurate. The result illustrates a common applied-ML failure mode: if a target is constructed from a physical quantity already present in the features, a high R^2 can reflect algebraic reconstruction rather than discovery or transferable engineering knowledge.

The residualized experiment strengthens this interpretation. Once the analytic speed component is removed, the available public control/powertrain channels fail to predict the remaining residual across unseen circuits. This does not imply braking, throttle, RPM or gear are physically unimportant. It means that, in this representation and with public telemetry resolution, they do not contribute a stable cross-circuit signal that improves prediction of the residual target.

The value of the work is therefore methodological and tooling-oriented: explicit distance alignment, transparent lap selection, within-driver comparisons, circuit-held-out evaluation, analytic baselines, failure analysis and a reproducible negative result. For a future predictive study, the target should be temporally or spatially

downstream from the input controls - for example, using braking-zone and turn-in features to predict later exit speed or subsequent time loss - rather than using features that directly define the target.

7. Limitations

- Adaptive segmentation is driver-specific. Segment IDs are not guaranteed to represent the same physical corner across drivers, especially at Silverstone and Suzuka. This version treats rows as local windows and does not make cross-driver segment-ID claims.
- The lap filter uses a 107% quick-lap threshold and no separate traffic/aborted-lap classifier. Lap deltas extend to 5.566 s, so some comparisons are not clean performance laps.
- Public FastF1 telemetry lacks steering angle, continuous brake pressure, tire temperatures/pressures, suspension channels, fuel load, setup parameters and other team-level data. Brake is Boolean.
- The fastest selected same-driver lap is an extreme-order reference and may embed favorable track evolution or noise; comparisons across qualifying phases can still contain session-evolution effects.
- Observations are clustered because many comparison laps share the same reference lap. Inferential p-values from row-level correlations are therefore omitted in this revision.
- The compact delta export is sampled at 50 m even though the analysis grid is 5 m; this naming and storage distinction must be documented in released code.
- The primary model and the earlier sensitivity audit used slightly different feature sets because `tyre_life_delta` was present in the headline model but absent from the audit baseline. This revision does not mix those metrics.
- The residual experiment is negative and should not be generalized to proprietary high-frequency team telemetry or other racing series.

8. Conclusion

Across 372 same-driver qualifying-lap comparisons and 2,487 adaptive segment observations from five 2024 Formula 1 circuits, a direct analytic inverse-speed reconstruction achieved mean leave-one-circuit-out MAE of 0.0408 s and mean R^2 of 0.8127, outperforming the best evaluated learned model (gradient boosting: 0.0563 s MAE, R^2 0.7258). After subtracting the analytic component, control and powertrain features failed to predict the residual across every held-out circuit (mean R^2 -0.275). The study therefore does not support the claim that machine learning provides independent cross-circuit predictive value for this target. Instead, it provides an engineering case study in telemetry decomposition, baseline design and methodological self-audit. The most important lesson is practical: before adding machine learning to a motorsport analysis problem, test whether the relevant physics or algebra already solves the task more accurately and transparently.

9. Data, Code and Reproducibility

The public release should include the extraction script, analysis script, exact dependency versions, session identifiers, derived feature tables, evaluation tables, baseline audit and figures. Raw Formula 1 telemetry should be re-fetched through FastF1 rather than redistributed without confirming upstream rights. The data source used for this study was FastF1 3.8.3. The extraction sessions were 2024 Bahrain Qualifying, Monaco Qualifying, British Grand Prix Qualifying, Italian Grand Prix Qualifying and Japanese Grand Prix Qualifying.

Primary analysis seed: 42. Reviewer-audit/residual check seed: 0. Model hyperparameters and selection rules are stated in Sections 2 and 4.

10. Research Integrity and AI-Assistance Disclosure

This is independent student research and is not presented as institutionally supervised or endorsed by New York Institute of Technology. Generative AI tools, including OpenAI ChatGPT and Anthropic Claude, were used

to assist with code development/debugging, manuscript drafting, and methodological review. The author executed the data extraction and analysis pipeline locally, supplied the resulting artifacts for audit, reviewed the reported outputs, and accepts responsibility for the final content, claims and disclosures. AI systems are not listed as authors. The manuscript was substantially revised after a critical review identified missing analytic baselines, target coupling, segment-consistency limitations and inconsistent feature-set reporting.

Author contribution: Kazi Sadia Anowar - project conception, local data acquisition, execution of the analysis pipeline, project integration, review of results, manuscript review and final accountability. AI-assisted contributions are disclosed above.

Conflict of interest: No conflict of interest is declared.

11. Revision Note

Version 2.0 replaces the original ML-first interpretation. It adds non-learned baselines; moves the residualized test into the main results; reports fold variation and target-scale normalization; corrects the Silverstone interpretation; defines the lap-selection rule; explains the 5 m analysis versus 50 m compact export; discloses the tyre-life feature-set mismatch; reports outlier ranges; and reframes adaptive segments as local windows rather than canonical cross-driver corner IDs.

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Appendix A. Key Reproducibility Numbers

- 469 selected laps; 467 successfully extracted telemetry laps; 372 comparison pairs; 2,487 adaptive segment observations.
- Original configured Gradient Boosting: MAE 0.0563 ± 0.0310 s across circuit folds; mean R^2 0.7258 ± 0.0389 .
- Analytic reconstruction: mean MAE 0.0408 s; mean R^2 0.8127.
- Residual model: mean held-out R^2 -0.275; negative on every circuit.
- Official vs integrated lap delta: $r=0.9937$ overall; mean absolute difference 0.0846 s; bias +0.0119 s; $r=0.9428$ for pairs under 1 s.
- 98.9% of lap pairs use the same tyre compound.